

TEMPLATE

KEY PROJECT INFORMATION & PROJECT DESIGN DOCUMENT (PDD)

PUBLICATION DATE **14.10.2020**

VERSION **v. 1.2**

RELATED SUPPORT

– **TEMPLATE GUIDE Key Project Information & Project Design Document v.1.2**

This document contains the following Sections

Key Project Information

SECTION A – Description of project

SECTION B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

SECTION C – Duration and crediting period

SECTION D – Summary of Safeguarding Principles and Gender Sensitive Assessment

SECTION E – Outcome of Stakeholder Consultations

Appendix 1 – Safeguarding Principles Assessment (mandatory)

Appendix 2 - Contact information of Project participants (mandatory)

Appendix 3 - LUF Additional Information (project specific)

Appendix 4 - Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	11328
Title of Project	HBRE Chu Prong Wind Power Farm Project
Time of First Submission Date	17/08/2021
Date of Design Certification	N/A
Version number of the PDD	2.1
Completion date of version	03/08/2023
Project Developer	Coral Future Pte. Ltd.
Project Representative	Coral Future Pte. Ltd.
Project Participants and any communities involved	HBRE Gia Lai Wind Power Joint Stock Company
Host Country (ies)	Viet Nam
Activity Requirements applied	☐ Community Services Activities ☒ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale ☒ Large Scale
Other Requirements applied	Not applicable
Methodology (ies) applied and version number	ACM0002: Grid-connected electricity generation from renewable sources - Version 21.0
Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☐ Regular ☒ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Climate change mitigation	147,519	tCO ₂ e
7 Affordable and Clean Energy	Energy generation, efficiency & access	169,900	MWh/year (of renewable energy generated)
8 Decent Work and Economic Growth:	Employment	30	Number of employees

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

HBRE Chu Prong Wind Power Farm Project involves the construction of an onshore wind power plant at Ia Bang and Ia Phin communes, Chu Prong district, Gia Lai province, Viet Nam. The main purpose of this project activity is to generate clean form of electricity through renewable wind energy sources by using wind turbines. The project activity installed 4x4.5MW and 8x4MW wind turbines with installed capacity of 50 MW and the estimated annual power generation of 169,900 MWh. The project activity will be connected to the national grid via 110 kV transformer.

The project boundary includes the wind project, sub-stations, grid and all power plants connected to the Vietnamese grid.

The baseline scenario of the project activity is the same as the scenario existing prior to the start of implementation of the project activity. The scenario is that the electricity would have been generated by the operation of grid-connected power plants which are mainly dominated by thermal/fossil fuel-based power plants and by the addition of new generation.

A.1.1. Eligibility of the project under Gold Standard

General Eligibility Criteria

As per section 3.1.1 of GS4GG Principles & Requirements, version 1.2, the project activity meets the eligible criteria as described below:

- (a) Types of Project: The project type is power generation using Wind Energy which is an eligible project type as it is in accordance with Renewable Energy Activity Requirements, version 1.4 (*refer to point h below*).
- (b) Location of Project: The project activity is located in Ia Bang and Ia Phin communes, Chu Prong district, Gia Lai province, Viet Nam.
- (c) Project Area, Project Boundary and Scale: The project activity follows the UNFCCC approved methodology and project area and boundary is defined in line with methodological requirement.

The proposed project will occupy 10.71 ha¹ land for powerhouse. The spatial extent of the project boundary includes the HBRE Chu Prong Wind Power Farm Project and all power plants/units connected physically to the electricity system that the proposed project is connected to. The details are described in section B.3 of the PDD.

The project's installed capacity is 50 MW and it thus qualifies under large scale projects. As per section 3.3.2 of GS4GG Renewable Energy Activity Requirements, version 1.4, This Renewable project estimates issuing emission reductions greater than 10,000 tCO₂e and its installed capacity is greater than 15 MW, thus the project falls under Non Microscale/ non small scale categories.

It is no potential for double counting for the project. Project area is for only construction of HBRE Chu Prong Wind Power Farm Project, and this project is seeking registration only in Gold Standard.

- (d) Host Country Requirements: Proposed project is in compliance with Viet Nam's legal, environmental, ecological and social regulation. The project owner has got all the required permits which granted by the government like as Business Certificate, Investment License of the Project, Certificate of Land Use Rights, Power Purchase Agreement (PPA), Approval of EIA.

Also, the project activity contributes to the national and local level sustainable development as follows:

Environmental sustainability

¹ Environmental Impact Assessment, 09/2019

- Through wind power utilization to generate electricity, the project will contribute to reduce in gases emissions (CO₂, NO_x, SO_x,...) including GHG emissions, which otherwise would have been generated by fossil fuels based power plants.
- Being a renewable energy source, wind energy used to generate electricity contributes to resource conservation and reduces reliance on exhaustible fossil fuel-based power sources as well as the need to import fuels for the purpose of power generation.

Economic sustainability

- With rapidly increasing demand of electricity consumption due to population growth and economic development in Viet Nam. The project will directly contribute towards balancing the supply and demand gap of electricity.

Moreover, the Gold Standard project will contribute to economic development of region through:

- The project activity will increase employment opportunities in the area where the Project is located, and then give an increase in the local community's income in general.
- The project activity will contribute to enhance the local investment environment, and thereby improves the local economy through the provision of stable power.
- The project activity contributes to diversifying the sources of electricity generation, important for meeting growing energy demands and the transition away from diesel and coal-supplied electricity generation.
- The project activity contributes to poverty alleviation through income and employment generation throughout project operation.
- The Project contributes towards the tax revenues of the province.

Social sustainability

- The proposed project will help to develop the area by providing substantial temporary job opportunities during the construction phase and stable and permanent job opportunities throughout the project operation for local residents.

Technology sustainability

- Modern and highly efficient turbines and generators will be used under this project, which in turn will accelerate the deployment of renewable energy

technologies in Vietnam.

- The project supports technological and know-how transfer from other regions or even other countries through training and practical work.

In conclusion the project will contribute positively towards sustainable development and be consistent with the energy policies set by the Government of Viet Nam.

- (e) Contact details: Project participants of the project activity include Coral Future Pte. Ltd. and HBRE Chu Prong Wind Power Farm Project.

Coral Future Pte. Ltd. has registration number 201906238H and has its registered office in Singapore.

HBRE Gia Lai Wind Power Joint Stock Company has registration number 5901071950 and registered office in Viet Nam. The contact details are provided in Appendix 2 of the PDD.

The project participants have operated in good status with no evidence of insolvency or legal/criminal notices placed against it or any of its directors.

- (f) Official Development Assistance (ODA) Declaration: No public funds or ODA are involved in the project activity.
- (g) Legal Ownership and Other Rights: HBRE Chu Prong Wind Power Farm Project is the project owner which has full legal right to control and operate the project activity.

The Coral Future is the project developer which is responsible for investment and management of the project development under the GS as well as the purchaser of the GS-VERs, and therefore The Coral Future has full and uncontested legal ownership of any GS-VERs that are generated under Gold Standard Certification through Environmental Attributes Purchase Agreement (EAPA) for the project.

The details are presented in section A.1.2 of this PDD.

The project activity meets the additional requirements as per section 2 and 3 of the "Renewable Energy Activity Requirement", version 1.4 as mentioned below:

- (h) Eligible project types

Section 2.1.2: The project shall generate and deliver electricity from renewable energy source of wind energy to the national grid. Thus, it meets this criterion.

Section 2.1.3: The project complies with the criteria as follows:

- The project is located in Viet Nam – a Lower Middle-Income country².
 - The penetration level – the ratio of installed capacity of the wind power of 963 MW³ (updated then end of August 2021) to the total grid installed capacity of 69,000 MW⁴ (end of 2020) is 1.19 %, which is less than 5%.
- (i) Suppressed demand: The Suppressed Demand baseline is only applied for small scale and microscale projects while this project activity is a large scale, so Suppressed Demand scenario was not applied for establishing the baseline.
- (j) The proposed project is not registered in any other voluntary or compliance schemes. The proposed project is seeking registration only in Gold Standard.
- (k) The proposed project is located in Viet Nam. Until date, there is no emission reduction caps or trade emissions in Viet Nam.
- (l) Double counting: No potential for double counting. Project area is for only construction of HBRE Chu Prong Wind Power Farm project, and this project is seeking registration only in Gold Standard. Therefore, there is no potential for double counting.
- (m) Stacking: As per section 3.5 of Renewable Energy Activity Requirement, version 1.4, GS VERs with REC labels (if any) cannot be claimed for the same MWh. The Project Activity will not claim both GS VERs and RECs for the same MWh. Because the project activity has not yet been registered under I-REC Standard until now. Hence stacking will be avoided.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

HBRE Gia Lai Wind Power Joint Stock Company is a project participant also as a project owner and they have the legal right to control and operate the project activity. The investment license, Certificate of Land use right are evidences for right of use. The approval of Feasibility Study Report (FSR) and Environmental Impact Assessment Report (EIAR) further support project owner's right. Besides, the PPA is the evidence for the ownership of the plant, equipment and process generation.

² <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups>

³ <https://moit.gov.vn/phat-trien-ben-vung/24-nha-may-dien-gio-voi-tong-cong-suat-la-963-mw-dua-vao-van-hanh-thuong-mai-den-cuoi-thang-8-2021.html>

⁴ [https://www.evn.com.vn/userfile/User/tcdl/files/2021/7/Ban_tin_EVN_so_7_nam_2021\(4_76_MB\).pdf](https://www.evn.com.vn/userfile/User/tcdl/files/2021/7/Ban_tin_EVN_so_7_nam_2021(4_76_MB).pdf)

Coral Future Pte. Ltd is the project developer and GS VER, RECs purchaser through Environmental Attributes Purchase Agreement for the project “HBRE Chu Prong Wind Power Farm Project”. They are also responsible for investment and management of the project development under Gold Standard Certification. Therefore, they have full legal ownership of any GS-VERs generated under GS certification or any RECs generated if applicable.

A.2. Location of project

The project is located in Ia Bang and Ia Phin communes, Chu Prong district, Gia Lai province, Viet Nam.

The geo-coordinate of the turbines that identify the boundary of the project are as follows:

Table 2: The geo-coordinate of turbines of the project activity⁵

No of wind turbine	Coordinate	
	Northern latitude (°)	Eastern longitude (°)
1	13.814359	107.981093
2	13.812354	107.982552
3	13.808781	107.982988
4	13.804046	107.982465
5	13.799535	107.977640
6	13.796878	107.978502
7	13.795654	107.982617
8	13.793027	107.982575
9	13.802634	107.971561
10	13.785965	107.965372
11	13.782790	107.966487
12	13.780084	107.966221

The site of the project area is shown as below figure:

⁵ FSR – page 5/159

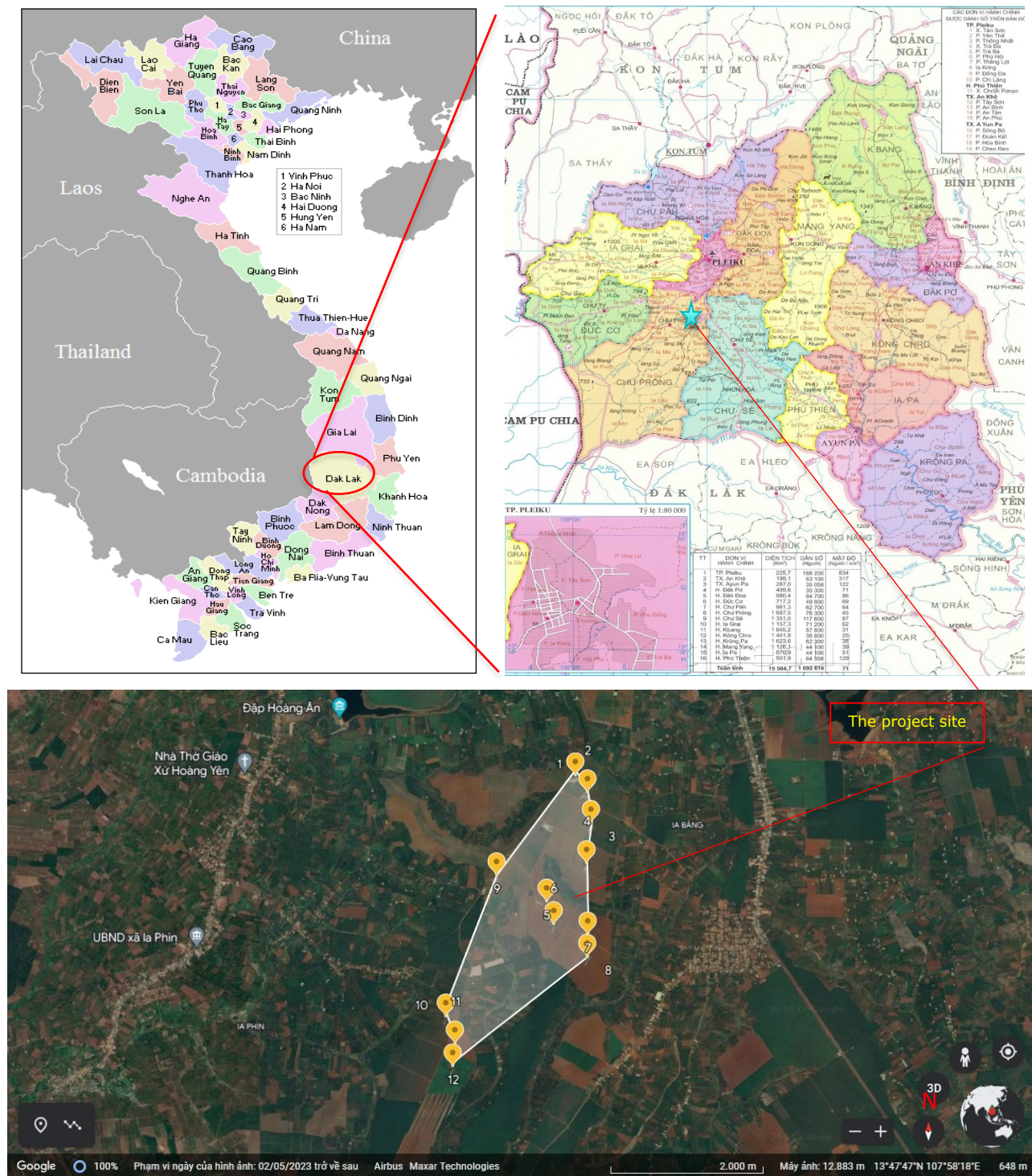


Figure 1. Map of the project site

A.3. Technologies and/or measures

The Project activity involves installation of one on-shore wind power project with installed capacity of 50 MW, containing 12 turbines and generators in order to convert

kinetic energy of the wind to electrical energy which is connected directly to national grid via 110 kV voltage level.

The HBRE Chu Prong wind power farm consists following main components:

- Turbine: Converts kinetic energy of the wind to kinetic energy of turbine.
- Generator: its function is to convert mechanical energy into alternating current electricity.
- Transformer: its purpose is to step up output voltage of generated electricity to voltage level of 110kV for connecting to national grid.

Technical specification of main equipment at the proposed project activity are shown in below table:

Table 3. Technical specification of main equipment⁶

Parameters	Value	
Turbine		
Type	Horizontal axis wind turbine	
Number	4	8
Model	GW 4.5MW-TFY	GW 4.0 MW-TFY
Rated power	4.5 MW	4.0 MW
Generator		
Type	Three phased	
Number	12	
Frequency	50 Hz	
Rated capacity	4800 kW	
Voltage	760 VAC	
Step-up transformers		
Type	SCLB11-50000-35	
Number	12	
Rated power	5000 kVA	
Voltage	690 kV/35 KV	
Frequency	50 Hz	
Main Transformer		
Type	Three-phased, out-door	
Number	1	
Voltage	35kV/110KV	

⁶ FSR – Technical Design Report, page 7/159

Rated output	63 MVA
Frequency	50 Hz
Plant load factor ⁷	38.79%
Technical lifetime of the plant ⁸	20 years

A.4. Scale of the project

The project activity is a renewable energy project with installed capacity of 50 MW which is larger than 15 MW, the project is therefore a large scale as per as per section 3.3 of Renewable Energy Activity Requirement, version 1.4.

A.5. Funding sources of project

No ODA funding is received for the proposed project activity

⁷ The Plant Load Factor (PLF) for this project was determined by the annual electricity output the capacity which were provided by the third party contracted by the project owner. So it is in line with para 3 (b) "The plant load factor determined by a third party contracted by the project participants (e.g. an engineering company)" of PLF guideline, Annex 11, EB48

⁸ FSR – Basic Design Report, page_122

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

Methodology:

- ACM0002: Grid-connected electricity generation from renewable sources - Version 21.0⁹

Tools referred with above methodology and applicable for project activity are:

- Tool07-Tool to calculate the emission factor for an electricity system¹⁰ - Version 07.0 (EB 100, Annex 4)
- Tool01 - Tool for the demonstration and assessment of additionality¹¹ - Version 07.0.0 (EB 70, Annex 8).
- Tool27 - Investment Analysis ¹²– Version 12.0 (EB116, Annex 2).
- Tool24 – Common Practice¹³ – Version 03.1 (EB 84, Annex 7).

B.2. Applicability of methodology (ies)

The project activity involves generation of grid connected electricity from renewable wind power energy. The project status is corresponding to the methodology ACM0002 version 21.0 and applicability of methodology is discussed below:

Table 4: Application of methodology to the proposed project activity

Applicability Criterion	Project Case
1. This methodology is applicable to grid connected renewable energy power generation project activities that: a) Install a Greenfield power plant; b) Involve a capacity addition to (an) existing plant(s); c) Involve a retrofit of (an) existing operating plants/units; d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or	The project activity is a renewable Energy Project i.e. wind power Project which falls under applicability criteria option 1 (a) i.e., "Install a Greenfield power plant". Hence the project activity meets the given applicability criterion.

⁹ <https://cdm.unfccc.int/methodologies/DB/HF3LP6O41YY0JIP1DK6ZRJO9RSCX3S>

¹⁰ <http://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-07-v7.0.pdf>

¹¹ <http://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-01-v7.0.0.pdf>

¹² <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-27-v12.pdf>

¹³ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-24-v1.pdf>

e) Involve a replacement of (an) existing plant(s)/unit(s)	
2. The methodology is applicable under the following conditions: a) The project activity may include renewable energy power plant/unit of one of the following types: hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	The option (a) of applicability criteria 2 is applicable as project is renewable energy wind power plant/unit.
3. In case of hydro power plants, one of the following conditions shall apply: a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4 W/m²; or d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4 W/m², all of the following conditions shall apply: i) The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; ii) Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; iii) Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a. Lower than or equal to 15 MW; and b. Less than 10 per cent of the total installed capacity of integrated hydro power project.	The project activity involves to construct a wind power plant, not a hydropower plant. Thus, the project activity is not applicable criteria 3.
4. In the case of integrated hydro power projects, project proponent shall:	The project is a wind power project and thus the criterion is not applicable.

a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum five years prior to implementation of CDM project activity.	
5. The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; b) Biomass fired power plants/units.	(a) The project activity is Greenfield and there is no switching of fossil fuel to renewable energy. Hence the criteria is not applicable to the project activity (b) The project is not a biomass fired power plant. Hence the criteria is not applicable to the project activity.
6. In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is "the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance".	Not applicable, this wind power project is a Green field project activity and this project is not the enhancement or up gradation project.

This comparison shows clearly that Version 21.0 of ACM0002 is applicable to the proposed project activity.

The project activity is also eligible to apply the methodological tool as listed above. Justification is discussed in below:

Table 5: Application of methodology tool to the proposed project activity

No.	Applicability	Project activities	In accordance with applicable conditions
Tool 07: Tool to calculate the emission factor for an electricity system, Version 07.0			

1	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g., demand-side energy efficiency projects).	The project activity is a greenfield wind power generation plant and hence, according to the applied methodology, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in "TOOL07: Tool to calculate the emission factor for an electricity system"	Yes
2	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, the conditions specified in "Appendix 2: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and	The project activity is grid connected wind power project, this condition is applicable, and the emission factor has been calculated by DNA Viet Nam accordingly.	Yes

	that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity		
3	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The project activity is located in Vietnam, which is a non-Annex I country. Therefore, this tool is applicable for the project activity	Not applicable
4	Under this tool, the value applied to the CO ₂ emission factor of bio fuels is zero	The project activity is a grid connected wind power project and therefore, this criterion is not applicable for the project activity	Not applicable
Tool 01: Tool for the demonstration and assessment of additionality, Version 07.0.0			
1	The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for project participants when proposing new methodologies. Project participants may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	The project participants do not propose a new methodology and the tool is used for the demonstration of additionality.	Yes
2	Once the additionally tool is included in an approved methodology, its application by project participants using this methodology is mandatory.	The tool is referred in the approved methodology and the project participant used the tool for the demonstration of additionality.	Yes

Tool 24: Common Practice, Version 03.1			
1	This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality	The project activity applies the methodological tool "Tool for the demonstration and assessment of additionality". Hence, the tool is applicable	Yes
Tool 27: Investment analysis, Version 12.0			
1	This methodological tool is applicable to project activities that apply the methodological tool "Tool for the demonstration and assessment of additionality", the methodological tool "Combined tool to identify the baseline scenario and demonstrate additionality", the guidelines "Non-binding best practice examples to demonstrate additionality for SSC project activities", or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	The project activity applies the methodological tool "Tool for the demonstration and assessment of additionality". Hence, the tool is applicable.	Yes

The tables above shows that this project activity is completely in accordance with the applicable conditions of the ACM0002 methodology - "Grid-connected electricity generation from renewable sources", version 21.0 and relevant tools.

B.3. Project boundary

As per ACM0002 version 21.0 - "The spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the CDM project power plant is connected to", the project boundary of the project activity is presented in the figure below:

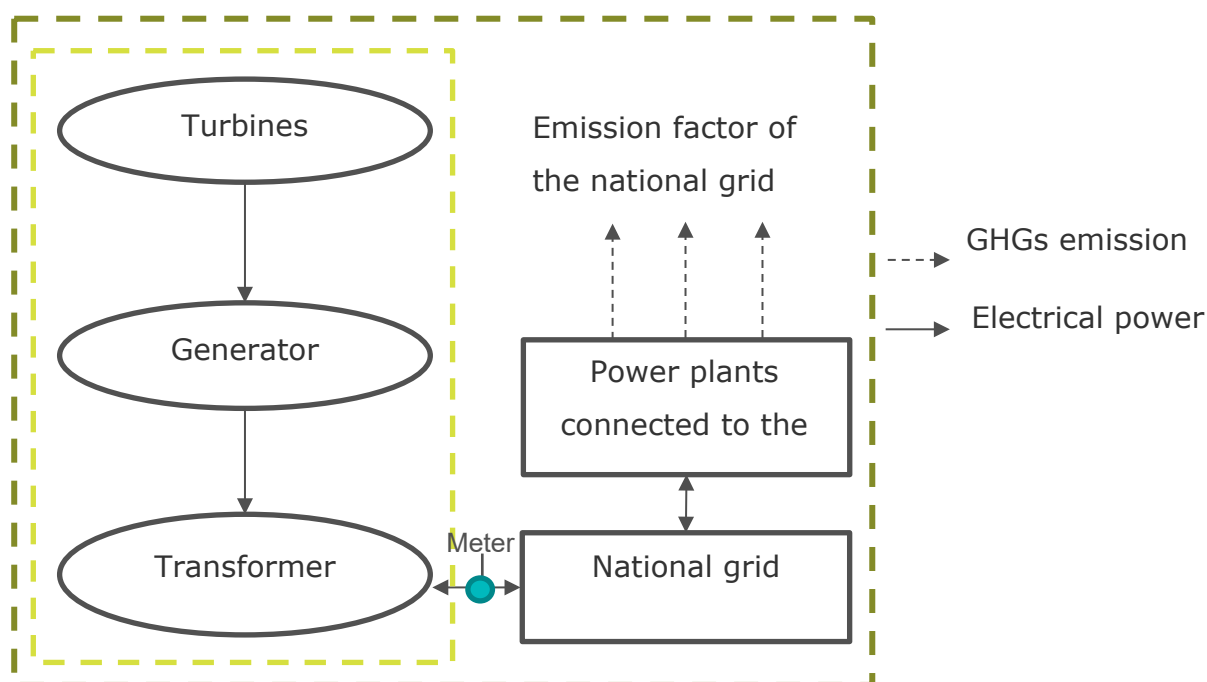


Figure 2. Project boundary

The GHGs and emission sources included or excluded from the project boundary are shown in Table below.

Table 6: Sources and gases included in or excluded from the project boundary

Source		GHGs	Included?	Justification/Explanation
Baseline scenario	CO2 emission from electricity generation fuel fired power plants that is displaced due to the project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Minor emission source
		N ₂ O	No	Minor emission source
Project scenario	For dry or flash steam geothermal power plants, emissions of CH4 and CO2 from non-condensable gases contained in geothermal steam	CO ₂	No	According to ACM0002, the project activity is a wind power plant and thereby all these sources are not applicable.
		CH ₄	No	
		N ₂ O	No	
	For binary geothermal power plants, fugitive emissions of CH4 and CO2 from non-condensable gases contained in geothermal steam	CO ₂	No	
		CH ₄	No	
		N ₂ O	No	
	For binary geothermal power plants, fugitive emissions of	CO ₂	No	

	hydrocarbons such as nbutane and isopentane (working fluid) contained in the heat exchangers	CH ₄	No
		N ₂ O	No
	CO ₂ emissions from combustion of fossil fuels for electricity generation in solar thermal power plants and geothermal power plants	CO ₂	No
		CH ₄	No
		N ₂ O	No
	For hydro power plants, emissions of CH ₄ from the reservoir	CO ₂	No
		CH ₄	No
		N ₂ O	No
	Charging of BESS using electricity from the grid or from fossil fuel electricity generators	CO ₂	No
		CH ₄	No
		N ₂ O	No

B.4. Establishment and description of baseline scenario

As per the approved consolidated Methodology ACM0002 (Version 21.0) para 24: "If the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the "Tool to calculate the emission factor for an electricity system".

The project activity involves setting up of a wind power project to produce electricity and supply to the grid. In the absence of the project activity, the equivalent amount of electricity would have been generated in the Vietnam grid by the operation of power plants dominated by fossil-based generation sources as well as by the operation of new power plants. Therefore, the CO₂ intensity of the power generation in the grid is calculated as the Combined Margin discussed below.

The combined margin ($EF_{grid,CM,y}$) is the result of a weighted average of two emission factor pertaining to the electricity system: the operating margin (OM) and the build margin (BM). Calculations for this combined margin must be based on data from an official source (where available) and made publicly available.

The combined margin of the Viet Nam grid ($EF_{grid,CM,y}$) used for the project activity is calculated in detail in section B.6.1 below.

B.5. Demonstration of additionality

According to version 21.0 of ACM0002, paragraph 31, the demonstration of additionality for the project activity is being carried out in accordance with the additionality tool provided by the UNFCCC i.e., “Tool01: Tool for demonstration and assessment of Additionality” Version 07.0.0.

The tool provides a step-wise approach to demonstrate additionality, which is displayed below:

Step 0: Demonstration whether the proposed project activity is the first-of its-kind

This step is not applied to the project activity since it is not first-of-its-kind, hence the additionality of the project will be demonstrated in next steps below.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity:

Paragraph 8 of version 07.0.0 of the additionality tool states: “Project activities that apply this tool in context of approved consolidated methodology ACM0002, only need to identify that there is at least one credible and feasible alternative that would be more attractive than the proposed project activity.”

We will therefore consider the two scenarios in the following analysis:

Alternative 1: The proposed project undertaken without the GS registration.

Alternative 2: Continuation of the current situation. In this case, the proposed project will not be constructed and the power will be solely supplied from the Vietnam national grid.

Sub-step 1b: Consistency with mandatory laws and regulations:

The alternative 2 “the continuation of the current situation” does not face with any barrier from the current law and regulation in Viet Nam because it is the “do-nothing” alternative. The project owner of a proposed project has no obligation to build or invest in the power plant to supply electricity for the local area. Hence this alternative is consistent with mandatory laws and regulations.

The alternative 1 is consistent with mandatory laws and regulations of Viet Nam as it was issued the Investment Certificate by People’s Committee of Gia Lai Province.

Both the alternatives enlisted above are found to comply with the mandatory laws and regulations taking into account the enforcement of the legislations in the region or country and EB decisions on national and/or sectoral policies and regulations.

However, alternative 2 has been selected as the appropriate baseline alternative for this project activity in line with the Tool01.

Step 2: Investment analysis

Determine whether the proposed project activity is not:

- (a) The most economically or financially attractive; or
- (b) Economically or financially feasible, without the revenue from the sale of emission reductions

To conduct the investment analysis, the following sub-steps are applied:

Sub-step 2a: Determine appropriate analysis method

Tool07 – “Tool for the demonstration and assessment of additionality”, Version 7.0.0 provides three methods of investment analysis: simple cost analysis (Option I), investment comparison analysis (Option II) and benchmark analysis (Option III).

The proposed project activity generates financial and economic benefits from sale of the generated electricity other than GS-VERs revenues, so the simple cost analysis (Option I) is not applicable. Out of the two remaining options, Option II is also not applicable as there are no other credible and realistic baseline scenario alternatives other than electricity supply from the grid. Thus, the benchmark analysis (Option III) is chosen to prove additionality.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using Benchmark analysis approach (Option III) has been chosen.

In the following, Project IRR is used to demonstrate the additionality of the project.

As indicated in paragraph 15, Tool 27 – “Investment analysis”, Version 12.0, “Local commercial lending rates or weighted average costs of capital (WACC) are appropriate benchmarks for a project IRR”, the project participant applies the local commercial lending rates as a benchmark for this project IRR. This benchmark is derived from the average long-term lending rates available from the beginning of calculated year up to the date of decision making. All data is sourced from weekly reports published by the State Bank of Vietnam on its official website (<https://www.sbv.gov.vn/en>).

The benchmark is 11% at the date of making the investment decision on 26 November 2019.

Sub-step 2c: Calculation and comparison of financial indicators

The key assumptions used to calculate the Pre-tax IRR of the proposed project are presented in the table below:

Table 3. Key assumptions to calculate the project IRR

No	Parameter	Unit	Value	Source
1	Gross capacity	MW	50	Feasibility Study Report (FSR)
2	Annual net electricity generation	MWh	169,900	FSR
3	Plant load factor	%	38.79	Calculated in IRR spreadsheet
4	Total investment cost	billion VND	1,810.694	FSR
	Unit cost per MW	Billion VND/MW	36.21	Calculated
5	Total annual O&M cost			FSR
	Year 1 (rate of construction and equipment cost)	%/ year	2.50	
	Year 2 - Year 5 (rate of construction and equipment cost)	%/ year	5	
	Year 6 - Year 20 (rate of construction and equipment cost)	%/ year	10	
6	Preparation period	year	1	FSR
7	Construction period	year	2	FSR
8	Period of financial assessment	year	20	FSR
9	Feed in tariff	VND/kWh	1,968.35	Calculated in IRR spreadsheet, based on exchange rate at investment decision making
		UScents/kWh	8.5	According to Decision ¹⁴ No.39/2018/QĐ-TTg on Amending several articles of Decision No.37/2011/QĐ-TTg dated June 29, 2011 of the Prime Minister on provision of assistance in development of wind power projects in Vietnam, the price for an on-shore wind power project is applied.
10	Residual/ salvage value	billion VND	0	According to Paragraph 6 of Tool27 - "Investment Analysis", Version 12.0, "Both project internal rate of return

¹⁴ <https://thuvienphapluat.vn/van-ban/EN/Dau-tu/Decision-39-2018-QĐ-TTg-amending-Decision-37-2011-QĐ-TTg-development-of-wind-power-projects/394945/tieng-anh.aspx>

				(IRR) and equity IRR calculations should reflect the period of expected operation of the underlying project activity (technical lifetime) and if a shorter period than the technical lifetime is chosen, the investment analysis shall be conducted for at least 10 years and include the fair value of the project activity assets at the end of the assessment period". Paragraph 7 of Tool27 - "Investment Analysis", Version 12.0, "The fair value should be calculated in accordance with local accounting regulations where available, or international best practice". Since the maximum depreciation period for wind projects according to Circular No.45/2013/TT-BTC issued by Ministry of Finance dated 25/04/2013 regarding "Guiding regulation on management, use and depreciation of fixed assets" is 20 years, which is equal the financial analysis period of 20 years for this project activity. Furthermore, the technical lifetime of 20 years has been applied as the financial analysis period, full value of assets has been completely depreciated thus no residual value remains at the end of the assessment period.
11	Pre-tax IRR	%	8.86	

The table shows that Project IRR of the proposed project was lower than the benchmark at the time of decision making which is defined as the date of issuing the decision to invest in the project by the Management Board on 26-11-2019.

Sub-step 2d: Sensitivity Analysis

A sensitivity analysis of the project activity has been conducted to test the robustness of the above calculations. Although the O&M cost accounted less than 20% of total investment cost that could be disregarded in the sensitivity analysis as regulated

under the guidance in Tool27 – “Investment analysis” – Version 09.0, paragraph 27, which states that “variables, including the initial investment cost, that constitute more than 20% of either total project costs or total project revenue should be subjected to reasonable variation”, this parameter is still considered in the analysis below. The following parameters are used in the sensitivity analysis of the project activity:

- Annual amount of electricity exported to the national grid
- Total investment cost
- Electricity price
- O & M cost

Table below shows the impact of variations in key factors on the Project IRR considering a $\pm 10\%$ variation in the parameters.

Table 4. Sensitivity analysis

No	Parameter	Variation	Project IRR	Likelihoods to happen
1	Annual amount of electricity exported to the national grid	+11.2 %	11%	The probability of a 11.2 % increase in annual export to the national grid is unlikely. This is because electricity generation is estimated based on large data series of the wind potential in the project area including wind speed, wind direction, temperature. Data has been monitored continuously each ten minutes in a year basis. Moreover, wind generation will degraded over its lifetime (It is found that level of degradation of wind farms may be $1.6 \pm 0.2\%$ of their output per year, (Ref. https://www.researchgate.net/)). Therefore, it is not possible to sustain a 11.2% increase.
		+10%	10.78%	Lower than the benchmark of 11%
		-10%	6.72%	Lower than the benchmark of 11%
2	Total investment cost	+10%	7.63%	Lower than the benchmark of 11%
		-10%	10.29%	Lower than the benchmark of 11%

		-14.4%	11%	The probability of a 14.4% decrease in the total investment cost is not likely to happen. Total investment cost is estimated by the third party, which is mentioned in the FSR and approved by Ministry of Industry and Trade. Furthermore, according to the General Statistics Office of Viet Nam, the Consumer Price Index (CPI) of housing and construction materials in Viet Nam show 102.50%; 104.27%; 103.32%; 103.03%, 101.79%, 101.76% in 2016, 2017, 2018, 2019, 2020 and 2021 respectively, which always increase compared to the previous year. While FSR was finalized in 05/2019 and the plant was operated in 2021, the actual value is 7.0% higher than the value in FSR. Therefore, this option shall be discarded. (Source: https://www.gso.gov.vn/)
3	Electricity price	+11.2%	11%	The probability of a 11.2% increase in feed in tariff annually is very unlikely because the PPA was signed with EVN with a fixed feed in tariff for long term. And in fact, the signed price in the PPA is 8.5 UScent/kWh and fixed for 20 years. Thus, this option shall be discarded.
		+10%	10.78%	Lower than the benchmark of 11%
		-10%	6.72%	Lower than the benchmark of 11%
4	O & M cost	+10%	8.31%	Lower than the benchmark of 11%
		-10%	9.37%	Lower than the benchmark of 11%
		-46.0%	11%	The probability of a -46.0% decrease in O&M cost to meet the benchmark is not possible to happen. According to General Statistics Office of Viet Nam, The Consumer Price Index (CPI) show an annual increase of 104.09%; 100.63%; 102.66%; 103.53%; 103.54%; 102.79%; 103.23%, 101.84% in 2014, 2015, 2016, 2017, 2018, 2019, 2020, and 2021 respectively in Viet Nam compared to the previous year. Inflation can be forecasted in the near future. The probability of such a decrease in operational cost is very small. Therefore, it is unlikely that the annual O&M cost will decrease by 46.0%. This option shall be discarded. (Source: https://www.gso.gov.vn/)

The sensitivity analysis shows that there is unlikely to be a scenario in which the variation of a parameter can make the project IRR without GS revenues reach the benchmark.

In conclusion, the proposed VCS project activity is unlikely to be financially attractive.

Step 3: Barrier analysis

Barrier analysis has not been applied.

Step 4: Common practice analysis

Sub-step 4a: Analyse other activities similar to the proposed project activity

Stepwise approach for common practice analysis has been carried out as per Methodological Tool24 - "Common Practice", Version 03.1.

Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

The proposed project activity has the installed capacity of 50 MW. So applicable capacity range as +/-50% is from 25 MW to 75 MW.

Step (2): Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- a) The projects are located in the applicable geographical area;
- b) The projects apply the same measure as the proposed project activity;
- c) The projects use the same energy source/fuel and feedstock as the proposed project activity, if a technology switch measure is implemented by the proposed project activity;
- d) The plants in which the projects are implemented produce goods or services with comparable quality, properties and applications areas (e.g. clinker) as the proposed project plant;
- e) The capacity or output of the projects is within the applicable capacity or output range calculated in Step 1;
- f) The projects started commercial operation before the project design document (CDM-PDD) is published for global stakeholder consultation or before the start date of proposed project activity whichever is earlier for the proposed project activity.

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

- a) The proposed project is located in Viet Nam so the entire host country (Viet Nam) was chosen as the applicable geographical area.
- b) The proposed project activity is a Greenfield wind power plant. Therefore, all the wind power plants (the same measure as the proposed project activity) are candidates for similar projects.
- c) The energy source used by the proposed project activity is wind. Hence, only wind power plants are candidates for similar projects.
- d) The proposed project activity produces electricity from wind energy to sell to EVN. Therefore, all the wind power plants connected to the national grid are candidates for similar projects

- e) The capacity range of the similar projects is within the applicable capacity range from 25 MW to 75 MW.
- f) The start date of the project under the Methodological tool for common practice is the date on which the project participants commit to making expenditures for the construction or modification of the main equipment or facility (e.g. a wind turbine), or for the provision or modification of a service (e.g. distribution of energy-efficient light bulbs, change of transport management system). Where a contract is signed for such expenditures (e.g. for procurement of a wind turbine), it is the date on which the contract is signed. In other cases, it is the date on which such expenditures are incurred. If the project activity involves more than one of such contracts or incurred expenditures, it is the first of the respective dates. Activities incurring minor pre-project expenses (e.g. feasibility studies, preliminary surveys) are not considered in the determination of the start date.

The Contract Agreement for Engineering, Procurement, Supply, Construction, Installation, Testing & Commissioning (EPC contract) among HBRE Gia Lai Wind Power Joint Stock Company and PowerChina Zhongnan Engineering Corporation Limited and PowerChina Vietnam Limited Company, is the first contract for expenditures of the proposed project activity was signed on 18/08/2020 and it is considered as the start date of the project activity for common practice analysis. The wind power projects in Viet Nam, which have started commercial operation before 18/08/2020 have been considered for analysis. These projects are listed as below:

Table 7. List of wind power projects considered in Viet Nam¹⁵

No.	Name	Capacity (MW)	Type	Commissioning year	Under carbon-credit scheme/ environmental credits	Investor
1	Wind Power No 1 - Binh Thuan	30	On-shore	2011	CDM ¹⁶	Vietnam Renewable Energy Joint Stock Company
2	Huong Linh 2	30	On-shore	09/2017	Not applicable	Tan Hoan Cau Corporation JSC ¹⁷
3	Huong Linh 1	30	On-shore	11/2019	Not applicable	Tan Hoan Cau Corporation JSC ¹⁸
4	Mui Dinh	37.6	On-shore	21/02/2019	REC ¹⁹	EAB Newenergy GmbH (Germany)

¹⁵ <https://devi-renewable.com/>

¹⁶ <https://cdm.unfccc.int/Projects/DB/KEMCO1219986182.6/view>

¹⁷ <https://www.power-technology.com/marketdata/power-plant-profile-huong-linh-2-wind-farm-vietnam/>

¹⁸ <https://www.power-technology.com/marketdata/power-plant-profile-huong-linh-i-vietnam/>

¹⁹ <https://evident.app/IREC/device-register/MUIDWIND001>

5	Dam Nai	37.8	On-shore	10/08/2017	REC ²⁰	The Blue Circle (Singapore) and TSV Viet Nam
6	Dai Phong	40	On-shore	24/07/2020	GCC ²¹	Dai Phong Development Investment Joint Stock Company
7	Tay Nguyen Wind – Phase 1	28.8		06/2019	Not applicable	HBRE Wind Energy Solutions Joint Stock Company ²²

Thus, numbers of similar projects identified is 7.

Step (3): Within the projects identified in Step (2), identify those that are neither registered CDM project activity, project activity submitted for registration, nor project activity undergoing validation. Note their number N_{all} .

In these above projects, 4 projects including Wind Power No 1 - Binh Thuan, Mui Dinh, Dam Nai, Dai Phong has been developed as carbon credits or environmental attribute certificates. So, $N_{all} = 3$.

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff} .

All of these projects in the step above are onshore wind power plants, which are the same to the proposed project. So, **$N_{diff} = 0$**

Step (5): Calculate factor $F=1-N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the proposed project activity that deliver the same output or capacity as the proposed project activity.

Calculate factor: $F = 1 - N_{diff}/N_{all} = 1$
 $N_{all} - N_{diff} = 3$

As per methodological tool “Common practice” - Version 03.1, the proposed project activity is a “common practice” within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all} - N_{diff}$ is greater than 3.

²⁰ <https://evident.app/IREC/device-register/DAMNWIND001>

²¹ <https://projects.globalcarboncouncil.com/project/329>

²² <https://autotimelapse.com/auto-time-lapse-cong-trinh-xay-dung/du-an-trang-trai-phong-dien-tay-nguyen/>

The above conditions are not fulfilled, $F = 1$ is greater than 0.2 but $N_{all} - N_{diff} = 3$ is equal to 3 (not greater than 3). Therefore, the project activity is not a “common practice” within a sector in the applicable geographical area, according to the guideline.

In conclusion, the proposed project activity is additional.

B.5.1. Prior Consideration

The project is a retroactive project so in order to eligible under Gold Standard, the required documents was submitted to Gold Standard for the first time on 17/08/2021 that is within one year of its start date. The following is the timeline for prior consideration:

No	Item	Timeline
1	Signing the Engineering Procurement and Construction contract	18/08/2020
2	Submitting documents for preliminary review	17/08/2021

B.5.2. Ongoing Financial Need

Not applicable

B.6. Sustainable Development Goals (SDG) outcomes

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)
13 Climate Action (mandatory)	Target 13.2: Integrate climate change measures into national policies and master plan of national grid strategies.	Climate change mitigation
		Reduction in GHGs emissions
7 Affordable and Clean Energy	7.2: By 2030, increase substantially the share of renewable energy in the global energy mix	Energy generation, efficiency & access
		Renewable energy generation
8 Decent Work and Economic Growth: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	Target 8.5: By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Employment
		Increased employment opportunities

B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

SDG Goal	Methodological choices/approaches
SDG 7 - Affordable and Clean Energy: Ensure access to affordable, reliable, sustainable and modern energy for all	This SDG outcome is measured in accordance with ACM0002 “Grid-connected electricity generation from renewable sources” version 21.0 for the net electricity supplied to the national grid by the project, $EG_{PJ,y}$ (MWh). Electricity imported ($EG_{import,y}$) and supplied ($EG_{export,y}$) to the grid will be measured through power meters. The net electricity will be calculated by subtracting $EG_{y,import}$ from $EG_{y,export}$ as follow: $EG_{PJ,y} = EG_{facility,y} = EG_{export,y} - EG_{import,y}$ Electricity exported to national grid is determined considering transmission loss as per "Technical design agreement of power metering system, data collection and measurement system of HBRE Chu Prong Wind Power Farm Project", No.5055/EPTC-KT&CNTT-KDBD dated 30/10/2019 between EVN and the Project Owner. Electricity exported to the national grid is calculated as multiplication of a conversion factor and electricity exported at metering point. <u>Baseline estimation approach</u> Prior implementation of the proposed project, there is no wind power plant installed in the project area, then no clean electricity is generated and supplied to grid. <u>Net estimation approach</u> The net benefit from project activity is estimated as follows: $\text{Net benefit} = \text{Project estimate} - \text{baseline estimate}$
SDG 8 – Decent Work and Economic Growth: Promote sustained, inclusive and sustainable economic growth, employment and decent work for all	<u>Project estimation approach</u> The project will create jobs during construction and operation process. Training and number of jobs will be monitored via training records, list of employees, salary level. During operation phase, about 30 long-term jobs will be provided for technicians qualified and unskilled workers. Refer: EIA, page 111 <u>Baseline estimation approach</u> Prior implementation of the proposed project, there is no wind power plant installed in the project area. Hence, no new jobs would have been created and no training would have been conducted.

	<u>Net estimation approach</u> The net benefit from project activity is estimated as follows: <u>Net benefit = Project estimate – baseline estimate</u>
SDG 13 – Climate Action : Take urgent action to combat climate change and its impacts	This SDG outcome is measured in accordance with ACM0002 “Grid-connected electricity generation from renewable sources” version 21.0 for the emission reductions generated by the project, ER_y (tCO₂e). Emission reduction is estimated as below: $ER_y = BE_y - PE_y$ Where: ER_y: Emission reduction (tCO₂e) BE_y: Baseline emission (tCO₂e) PE_y: Project emission (tCO₂e) <i>Baseline emission</i> As per para 47 and 49 of the applied methodology, baseline emissions is calculated as follows: $BE_y = EG_{PJ,y} * EF_{grid,CM,y}$ Where: BE_y Baseline emissions in year y (tCO₂/yr). $EG_{PJ,y}$ Quantity of net electricity generation delivered to national grid in year y (MWh/yr) $EF_{grid,CM,y}$ Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system”, version 7.0. The latest Emission factor of Viet Nam national grid is calculated and published by DNA Viet Nam, Department of Climate Change, Ministry of Natural Resources and Environment on 26/02/2021 (Official Letter No. 116/BDKH-TTBVTOD dated 26 February 2021²³, including: - Operating margin emission factor : $EF_{grid,OM,y} = 0.8907 \text{ tCO}_2/\text{MWh}$; and - Build margin emission factor : $EF_{grid,BM,y} = 0.8010 \text{ tCO}_2/\text{MWh}$

²³ [http://www.dcc.gov.vn/van-ban-phap-luat/1067/Nghien-cuu,-xay-dung-he-so-phat-thai-\(EF\)-cua-luoi-dien-Viet-Nam-nam-2019-\(k%C3%A8m-CV-116/BDKH-TTBVTOD\).html](http://www.dcc.gov.vn/van-ban-phap-luat/1067/Nghien-cuu,-xay-dung-he-so-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-nam-2019-(k%C3%A8m-CV-116/BDKH-TTBVTOD).html)

	The combined margin emissions factor is calculated as follows: $EF_{grid,CM,y} = EF_{grid,OM,y} \times W_{OM} + EF_{grid,BM,y} \times W_{BM}$ In which: <table> <tr> <td>$EF_{grid,CM,y}$</td><td>Combined margin CO₂ emission factor of national grid in year y (tCO₂/MWh)</td></tr> <tr> <td>$EF_{grid,OM,y}$</td><td>Operating margin CO₂ emission factor of national grid in year y (tCO₂/MWh)</td></tr> <tr> <td>W_{OM}</td><td>Weighting of operating margin emissions factor (%)</td></tr> <tr> <td>$EF_{grid,BM,y}$</td><td>Build margin CO₂ emission factor of national grid in year y (tCO₂/MWh)</td></tr> <tr> <td>W_{BM}</td><td>Weighting of build margin emissions factor (%)</td></tr> </table> The report of the DNA of Vietnam applying the weighting of margin emission factor and build margin emission factor are 0.5, these weightings are not applicable to wind power plants. According to version 07.0 of the "Tool to calculate the emission factor for an electricity system", the following default weightings are applied to wind power plants: $W_{OM} = 0.75 \text{ and } W_{BM} = 0.25$ Therefore, the combined margin emissions factor for wind power plants is calculated as follows: $EF_{grid,CM,y} = 0.75 \times 0.8907 + 0.25 \times 0.8010 = 0.868275$ (tCO₂/MWh) <i>Project emission</i> The project emission for the wind project is: $PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$ Where: <table> <tr> <td>PE_y</td><td>Project emissions in year y (tCO₂e/yr)</td></tr> <tr> <td>$PE_{FF,y}$</td><td>Project emissions from fossil fuel consumption in year y (tCO₂/yr)</td></tr> <tr> <td>$PE_{GP,y}$</td><td>Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (tCO₂e/yr)</td></tr> <tr> <td>$PE_{HP,y}$</td><td>Project emissions from water reservoirs of hydro power plants in year y (tCO₂e/yr)</td></tr> </table> The proposed project is a wind power plant that neither uses fossil fuel nor operates geothermal power plants or having water reservoirs (i.e. $PE_{FF,y} = 0$; $PE_{GP,y} = 0$; $PE_{HP,y} = 0$); therefore, the project emission is zero: $PE_y = 0$	$EF_{grid,CM,y}$	Combined margin CO ₂ emission factor of national grid in year y (tCO ₂ /MWh)	$EF_{grid,OM,y}$	Operating margin CO ₂ emission factor of national grid in year y (tCO ₂ /MWh)	W_{OM}	Weighting of operating margin emissions factor (%)	$EF_{grid,BM,y}$	Build margin CO ₂ emission factor of national grid in year y (tCO ₂ /MWh)	W_{BM}	Weighting of build margin emissions factor (%)	PE_y	Project emissions in year y (tCO ₂ e/yr)	$PE_{FF,y}$	Project emissions from fossil fuel consumption in year y (tCO ₂ /yr)	$PE_{GP,y}$	Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (tCO ₂ e/yr)	$PE_{HP,y}$	Project emissions from water reservoirs of hydro power plants in year y (tCO ₂ e/yr)
$EF_{grid,CM,y}$	Combined margin CO ₂ emission factor of national grid in year y (tCO ₂ /MWh)																		
$EF_{grid,OM,y}$	Operating margin CO ₂ emission factor of national grid in year y (tCO ₂ /MWh)																		
W_{OM}	Weighting of operating margin emissions factor (%)																		
$EF_{grid,BM,y}$	Build margin CO ₂ emission factor of national grid in year y (tCO ₂ /MWh)																		
W_{BM}	Weighting of build margin emissions factor (%)																		
PE_y	Project emissions in year y (tCO ₂ e/yr)																		
$PE_{FF,y}$	Project emissions from fossil fuel consumption in year y (tCO ₂ /yr)																		
$PE_{GP,y}$	Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (tCO ₂ e/yr)																		
$PE_{HP,y}$	Project emissions from water reservoirs of hydro power plants in year y (tCO ₂ e/yr)																		

	<u>Leakage emission</u> No other leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected.
--	--

B.6.2. Data and parameters fixed ex ante

SDG13

Data/parameter	$EF_{grid,OM,y}$
Unit	tCO ₂ /MWh
Description	Operating margin CO ₂ Emission Factor of national electricity grid in year y
Source of data	DNA of Viet Nam ²⁴
Value(s) applied	0.8907
Choice of data or Measurement methods and procedures	The latest $EF_{grid,OM,y}$ for 2019 was calculated and published by Ministry of Natural Resources and Environment, Department of Climate Change (DNA of Viet Nam) on 26/02/2021 as per "Tool to calculate the emission factor for an electricity system", version 07.0
Purpose of data	For calculation of the baseline emission
Additional comment	This parameter is fixed ex-ante for entire crediting period

Data/parameter	$EF_{grid,BM,y}$
Unit	tCO ₂ /MWh

²⁴ [http://www.dcc.gov.vn/van-ban-phap-luat/1067/Nghien-cuu,-xay-dung-he-so-phat-thai-\(EF\)-cua-luoi-dien-Viet-Nam-nam-2019-\(k%C3%A8m-CV-116/BDKH-TTBVTOD\).html](http://www.dcc.gov.vn/van-ban-phap-luat/1067/Nghien-cuu,-xay-dung-he-so-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-nam-2019-(k%C3%A8m-CV-116/BDKH-TTBVTOD).html)

Description	Build margin CO ₂ Emission Factor of national electricity grid in year y
Source of data	DNA of Viet Nam
Value(s) applied	0.8010
Choice of data or Measurement methods and procedures	The latest EF _{grid,OM,y} for 2019 was calculated and published by Ministry of Natural Resources and Environment, Department of Climate Change (DNA of Viet Nam) on 26/02/2021 ²⁵ as per "Tool to calculate the emission factor for an electricity system", version 07.0
Purpose of data	For calculation of the baseline emission
Additional comment	This parameter is fixed ex-ante for entire crediting period

Data/parameter	EF _{grid,CM,y}
Unit	tCO ₂ /MWh
Description	Combined margin CO ₂ Emission Factor of national grid applying for wind power plant in year y
Source of data	DNA of Viet Nam
Value(s) applied	0.868275
Choice of data or Measurement methods and procedures	The EF _{grid,CM,y} is calculated using EF _{grid,OM,y} and EF _{grid,BM,y} which was published by DNA of Viet Nam and weightings w _{OM} , w _{BM} of 0.75 and 0.25 for wind power plant project as per "Tool to calculate the emission factor for an electricity system" – Version 07.0
Purpose of data	For calculation of the baseline emission
Additional comment	This parameter is fixed ex-ante for entire crediting period

²⁵ [http://www.dcc.gov.vn/van-ban-phap-luat/1067/Nghien-cuu,-xay-dung-he-so-phat-thai-\(EF\)-cua-luoi-dien-Viet-Nam-nam-2019-\(k%C3%A8m-CV-116/BDKH-TTBVTOD\).html](http://www.dcc.gov.vn/van-ban-phap-luat/1067/Nghien-cuu,-xay-dung-he-so-phat-thai-(EF)-cua-luoi-dien-Viet-Nam-nam-2019-(k%C3%A8m-CV-116/BDKH-TTBVTOD).html)

B.6.3. Ex ante estimation of SDG Impact

SDG 7 - Affordable and Clean Energy

Project estimate

Net electricity generated by the project is calculated as the difference between quantity of electricity supplied to the national grid by the project ($EG_{\text{export},y}$ (MWh)) and amount of electricity imported from the grid ($EG_{\text{import},y}$ (MWh)) in accordance with ACM 0002 "Grid-connected electricity generation from renewable sources" version 21.0 for the net electricity supplied to the national grid by the project, $EG_{PJ,y}$ (MWh).

For the project activities include the installation of a new renewable energy power plant, $EG_{PJ,y}$ is calculated as follow:

$$EG_{PJ,y} = EG_{\text{facility},y} = EG_{\text{export},y} - EG_{\text{import},y}$$

In which:

- $EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the GS project activity in year y (MWh/yr)
- $EG_{\text{facility},y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)
- $EG_{\text{export},y}$ = Quantity of electricity supplied to the grid by project activities in the year y (MWh/yr)
- $EG_{\text{import},y}$ = The amount of electricity imported from the grid by the project activity at night or when the plant has a problem in the year y (MWh/yr)

Quantity of net electricity provided by the project activity to the national grid in the year is:

$$EG_{PJ,y} = 169,900 \text{ MWh/yr}$$

Baseline estimate

No wind power plant has been installed in the project area prior implementation of the proposed project, then no clean electricity is generated and supplied to grid. Hence:

$$\text{Baseline estimate} = 0 \text{ (MWh/yr)}$$

Net benefit

$$\text{Net benefit} = \text{Project estimate} - \text{Baseline estimate} = 169,900 \text{ (MWh/yr)}$$

SDG 8 - Decent Work and Economic Growth

Project estimate

The project will lead to employment opportunities which would not have been possible in the baseline scenario. The project is expected to provide permanent employment at around 30 persons (*Refer: EIA, page 111*). Also project activity improves the quality of employment by giving training to employees.

Baseline estimation approach

No wind power plant installed in the project area prior implementation of the proposed project. Hence, no new jobs would have been created and no training would have been conducted, and Baseline estimate is considered as zero.

Net estimation approach

The net benefit from project activity is estimated as follows:

$$\text{Net benefit} = \text{Project estimate} - \text{baseline estimate} = 30 \text{ jobs}$$

SDG 13 – Climate Action

Emission reduction of the project activity estimates as following equation:

$$ER_y = BE_y - PE_y$$

Where:

ER_y : Emission reduction (tCO₂e)

BE_y : Baseline emission (tCO₂e)

PE_y : Project emission (tCO₂e)

Baseline emission

Baseline emissions is calculated as follows:

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y}$$

Where:

BE_y Baseline emissions in year y (tCO₂/yr).

$EG_{PJ,y}$ Quantity of net electricity generation delivered to national grid in year y (MWh/yr)

$EF_{grid,CM,y}$ Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system", version 7.0.

$$BE_y = EG_{PJ,y} * EF_{grid,CM,y} = 169,900 * 0.868275 = 147,519 \text{ (tCO}_2\text{e/yr) (round down)}$$

Project emission (PE_y)

As mentioned above, $PE_y = 0$.

Leakage emission (LE_y)

As mentioned above, $LE_y = 0$.

Thus, $ER_y = BE_y = 147,519$ (tCO₂e/yr)

B.6.4. Summary of ex ante estimates of each SDG Impact

SDG 7 – Ensure access to affordable, reliable, sustainable and modern energy for all

Table 8: Net electricity supplied to national grid

Year	Baseline estimate	Project estimate	Net benefit
2021 (29/10-31/12/2021)	0	29,791	29,791
2022	0	169,900	169,900
2023	0	169,900	169,900
2024	0	169,900	169,900
2025	0	169,900	169,900
2026 (01/01 – 28/10/2026)	0	140,109	140,109
Total	0	849,500	849,500
Total number of crediting years	5		
Annual average over the crediting period	0	169,900	169,900

SDG 8 - Promote inclusive and sustainable economic growth, employment and decent work for all

The project is expected to provide permanent employment at around 40 persons. (Refer: EIA, page 111). Also annually project activity will improve the quality of employment by giving training to employees.

SDG 13 - Take urgent action to combat climate change and its impacts

Table 9: Expected emission reduction

Year	Baseline estimate	Project estimate	Net benefit
2021 (29/10-31/12/2021)	25,866	0	25,866

2022	147,519		147,519
2023	147,519	0	147,519
2024	147,519	0	147,519
2025	147,519	0	147,519
2026 (01/01 – 28/10/2026)	121,653	0	121,653
Total	737,595	0	737,595
Total number of crediting years	5		
Annual average over the crediting period	147,519	0	147,519

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 7 – Ensure access to affordable, reliable, sustainable and modern energy for all

Data / Parameter	$EG_{PJ,y}$
Unit	MWh/yr
Description	Quantity of net electricity of the project is exported to the national grid in year y
Source of data	Calculated based on measurement from bi-directional power meters (main and back-up meters) that are used for measuring the following parameters: - The quantity of electricity exported by the project plant to the grid, ($EG_{export,y}$ (MWh)); and - The quantity of electricity delivered to the project plant from the grid, ($EG_{import,y}$ (MWh)).
Value(s) applied	169,900
Measurement methods and procedures	Calculated as the power output of the power plant to the national grid minus the electricity input that the project receives from the national grid.

	$EG_{PJ,y} = EG_{facility,y} = EG_{export,y} - EG_{import,y}$
Monitoring frequency	Continuous measurement and at least monthly recording
QA/QC procedures	Power meters will be calibrated periodically according to national standards and regulations. As per DLVN 39:2019 – “Power meters – Verification/calibration procedure”, the calibration and verification for 3 phase meters need to be conducted every three years by third party. Data on electricity export and import will be cross-checked with a summary table on electricity delivery and receiving monthly and invoices
Purpose of data	Used for calculating baseline emission
Additional comment	N/A

SDG 8 - Promote inclusive and sustainable economic growth, employment and decent work for all

Data / Parameter	Number of jobs
Unit	Number of employees
Description	The project will create employment opportunities, involving various jobs for technicians, qualified and unskilled workers. Labor contract is made in accordance with host country laws.
Source of data	Labor contract, training records, salary records/ interview with Project owner
Value(s) applied	Number of employees: 30 (expected in operation phase) Average income: higher than local average income level 100% of permanents employees will be provided labor contracts, social and medical insurance
Measurement methods and procedures	Checking document provided by Project Participants.
Monitoring frequency	Once per monitoring period
QA/QC procedures	Cross-checking by interview
Purpose of data	To monitor the SDG 8 indicator
Additional comment	N/A

SDG 13 - Take urgent action to combat climate change and its impacts

Data / Parameter	ER _y
Unit	tCO ₂ e/year
Description	The project will generate renewable power, which will displace part of the electricity otherwise supplied by fossil fuel fired power plants, thus GHG emission reductions are achieved via the proposed project.
Source of data	Calculated
Value(s) applied	147,519
Measurement methods and procedures	Calculated according to methodology ACM0002, version 21.0
Monitoring frequency	Once per monitoring period
QA/QC procedures	Cross-checking with electricity data
Purpose of data	Direct emission reductions of the Project over the Baseline
Additional comment	N/A

Safeguarding Principle Assessment

Safeguarding principle Data / Parameter	9.5 Hazardous and Non-hazardous waste
Unit	N/A
Description	Non-hazardous wastes and hazardous wastes (if any) are collected and treated by the authorized third party in accordance with local laws and related regulations
Source of data	Site observation, hazardous treatment contract or non-hazardous treatment agreement.
Value(s) applied	N/A
Measurement methods and procedures	Site observation and checking documents
Monitoring frequency	Once per monitoring period

QA/QC procedures	Cross-checking by interview
Purpose of data	Prevent from penetration pollutants into the environment.
Additional comment	N/A

B.7.2. Sampling plan

Not applicable

B.7.3. Other elements of monitoring plan

Project Participants proposed the following structure for monitoring, data collection, data archiving and calibration of equipment for this project activity.

Organizational structure, roles and responsibilities of personnel

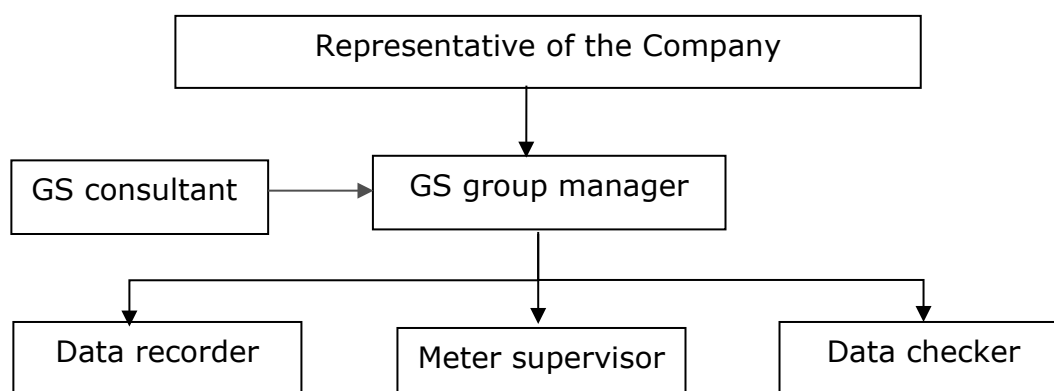


Figure 3. Structure of the monitoring group

Table 10. Group members and their responsibilities

Person	Responsibility
Representative of the Company	Review the monitoring report annually.
GS group manager	Managing the whole GS business of the plant, guiding and supervising data recorder
Data recorder	Collecting and recording data monthly.
Meter supervisor	Checking power meter periodically according to relevant regulation.
Data checker	Double checking the collected data measured by power meter.
GS consultant	Provide GS monitoring instruction and prepare reports to Sustain Cert.

Data management

The electricity generated will be monitored at the main meters and the back-up meters at delivery point at the respective HBRE Chu Prong Wind Power Farm project. The metering system will include one main meter and the two back-up meters according to "Technical design agreement of power metering system, data collection and measurement system of HBRE Chu Prong Wind Power Farm Project", No.5055/EPTC-KT&CNTT-KDBD dated 30/10/2019 between EVN and the Project Owner.

The main and backup power meters are used in order to record the electricity generated to the grid.

- In normal conditions, the data of main power meter will be used as the basis of electricity calculation for payment by the EVN to the PO and to calculate the emission reductions by the project activity.
- In case the main power meter is in failure and the backup power meter is still in good operation, the data of the backup meter 1 will be used to calculate electricity exported and imported and the emission reduction by the project activity.
- In case of both main and backup power meter 1 are in failure, the data of the backup meter 2 will be used to calculate electricity exported and imported and the emission reduction by the project activity.
- In case of all power meters are not operated, the Project Owner and the power company (EVN) will jointly calculate a conservative estimate of power supplied to the grid based on historical data. The assumptions used to estimate net electricity supply to the grid will be signed by both a representative of the project owner as well as a representative of the power company (EVN).

In case any power meters are in failure, the Project Owner will inform EVN immediately and contract with the authorized party (belong to EVN) to verify/ calibrate and/or replace the failed power meter.

Technical specification of power meters will be as follows:

Parameter	Main meter	Back-up meters	
	CTC1	CTP1	CTP2
Serial no.	20015978	21000348	21000342
Type	3 phases, 4 wires	3 phases, 4 wires	3 phases, 4 wires
Model	Ester-A1700	Ester-A1700	Ester-A1700
Accuracy	0.2s	0.5s	0.5s
Calibration	Calibrated	Calibrated	Calibrated

The monitoring system diagram of the proposed project activity is shown as below Figure:

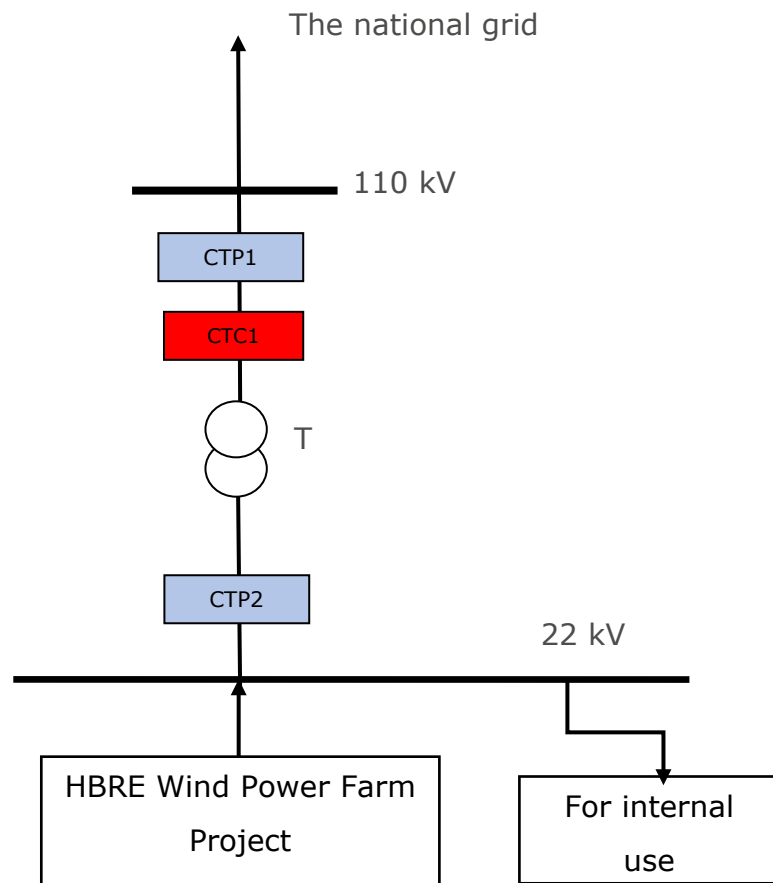


Figure 4. Connection diagram

Where:

CTC1: Main power meter

CTP1, CTP2: Back-up power meters

T: Transformer

Data collection and archiving:

- The GS group appointed by the wind power plant shall keep monitored data in electronic archives at the end of every month. Paper documents should be stored in electronic format. Electronic documents should be printed out and kept.
- The wind power plant shall keep the copy of electricity sale invoices (the original electricity sale invoices shall be kept by Finance Department of the Company).
- All the data and information in the form of paper documents shall be archived by the GS group, with at least one copy backup for each datum.

- Data monitored and required for verification and issuance are to be kept for two years after the end of the crediting period or the last issuance of GS-VERs for this project activity, whichever occurs later.

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

18/08/2020 (The earliest date on which the Project Owner has committed to expenditures related to the implementation of the Project, i.e the signed date of EPC Contract)

The project activity is a retroactive project once stakeholder consultation has been conducted on 11/04/2022 after the project start date.

C.1.2. Expected operational lifetime of project

20 years 0 month

C.2. Crediting period of project

C.2.1. Start date of crediting period

29/10/2021 (the start date of commercial operation)

C.2.2. Total length of crediting period

5 years 0 month

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarized below.

Principles	Mitigation Measures added to the Monitoring Plan
Principle 9.5 Hazardous and Non-hazardous Waste	Refer to B7.1

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project participants do not involve and promote any discrimination based on gender. Both men and women were invited and participated in the stakeholder consultation meeting. Both men and women labours are employed and arranged suitable positions. The project being a renewable energy project is not gender sensitive project. The project does not adversely impact women or men.
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The project activity is aligned with the Gender Equality Law approved by the National Assembly in 2006 and the Government's Decree No. 48/2009/ND-CP providing measures to assure gender equality in Viet Nam.
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	No, Expert is not required. The gender equality is addressed fully in the Safeguarding Principles assessment in Appendix 1.
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	No, Expert is not required. The project being a renewable energy project is not gender sensitive project. And in fact, no gender related issues were raised at the Stakeholder Consultation.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

Due to COVID-19 outbreak, the integrated physical meeting for local stakeholder consultation and stakeholder feedback round is postponed, and the draft PDD and draft stakeholder consultation report was submitted for preliminary review. The meeting was conducted on 11 April 2022 as the Covid -19 situation is less stressful in Viet Nam. Please refer to the separate Stakeholder Consultation Report for more information..

E.1. Summary of stakeholder mitigation measures

Since the project is retroactive, one consultation with a physical meeting integrated into the SFR was conducted on 11 April 2022. During the consultation, all participants agreed that negative impacts are insignificant, and proposed mitigation measures from Project owner are proper. There are no comments concerned about environmental, social, and

economic concerns, which caused the addition of mitigation measures. There is no mitigation measure added by the stakeholders.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	Comment Books were made available at Project site and at offices of other Project participants in order to stakeholders can provide feedback on the proposed project.
GS Contact (mandatory)	help@goldstandard.org
Other	The telephone numbers of the Project Owner and other Project Participants are made available for local stakeholders to provide feedback on the project. Stakeholders can find the telephone number in the Comment Book at the project site and the office of other Project Participants: - HBRE Gia Lai Wind Power Joint Stock Company Address: 282 Hung Vuong street, Chu Prong town, Chu Prong district, Gia Lai province, Viet Nam. Contact details: Nguyen Hong Hanh Mobile: +84 979728019/ Email: tuvannangluongxanh@gmail.com. - Coral Future Pte. Ltd Address: 20A, Tanjong Pagar Road, Singapore, 088443 Contact details: Ms. Tran Thi Huong Mobile: ++84 979728019/ Email: huongtt.vngecc@gmail.com

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion	1. No 2. No	1. Viet Nam ratified International Convention on the Elimination of all Form of Racial Discrimination" on 09/06/1981; "International Covenant on Civil and Political Rights" and "International Covenant on Economic, Social and Cultural Rights" on 24/09/1982; "Convention on the Elimination of all Forms of Discrimination against Women" on 27/11/1981. Therefore, the project developer and the project do respect nationally and internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind 2. Viet Nam ratified International Convention on the Elimination of	1. N/A 2. N/A

		all Form of Racial Discrimination” on 09/06/1981; “Convention on the Elimination of all Forms of Discrimination against Women” on 27/11/1981. Therefore, the project will not discriminate with regards to participation and inclusion.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country’s national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)	1. No 2. No 3. No 4. No	1. The project owner have to comply with the Labour Code in Viet Nam law (Chapter X – Separate provision for female employees). The project activity is to supply clean energy from the wind power plant to the national grid. Thus it does not involve any forms of violence against women or lead to/contribute to adverse impacts on gender equality. The project does not reduce or put at risk women’s access to or control of resources, entitlements and benefits. 2. The project owner complies with regulation in Vietnam’s Labour Code. Trained labours are used, local people are engaged for the same for suitable jobs. No	1. N/A 2. N/A 3. N/A 4. N/A

		discrimination either in gender, or any other forms is occurred to employees. Employees are paid equally for equal work. 3. The project does not have any scope to apply gender strategy as such. Although the project positively contributes towards the national mission for empowerment of women through improvement of health and attaining vision for empowerment of women under Law on Gender Equality 2006 4. N/A	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community	1. Yes	The project leads to safe and improvement in health of the workers and the community as it will replace coal as fuel with wind to generate clean and safe electricity. Further, periodic maintenance by implementing agency ensure prevention of any unsafe working condition. All employees will attend health & safety trainings or fire protection and prevention trainings.	1. Training on Health and safety will be conducted regularly during project operation. This is a monitored parameter under section B.7.1.
Principle 4.1 Sites of Cultural and Historical Heritage			

Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project activity will occupy approximately 10.71 ha area, which is a hilly area with rubber trees, undulating terrain. There are no people living there so it is advantage for developing a wind power project. The project site is not included any works; any cultural or historical heritage. <i>Refer: EIA Report, page 11, 13.</i>	N/A
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project site is no people living there. There is no migration and resettlement. The project thus doesn't require or cause the physical or economic relocation of people. <i>Ref. EIA Report, page 13</i> The occupied land includes mainly rubber plantation, which is managed by private company, and a small part of residents. <i>Ref. EIA Report, page 80, 81</i> The project owner implemented negotiation, compensation and supports with relevant entities in line with Viet Nam's regulation. <i>Ref. Approval of compensation plan; Compensation documents</i>	N/A
Principle 4.3 Land Tenure and Other Rights			

Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership?	No	A land lease agreement between People's Committee of Gia Lai province and HBRE Gia Lai Wind Power JSC (PO) was signed. Thus, there is no uncertainty related to land tenure arrangements and/or access rights, usage rights or land ownership. An energy transmission line with dual circuit and the length of 1.4km will be built to connect from 110kV substation at the plant to national grid. The land area occupied for building electricity transmission line is 0.315 ha.	N/A
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects	1. No	1. There is no corruption provision in the project activity	1. N/A
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the	1. No 2. No 3. No 4. No 5. No	1. Forced labour is an illegal activity in Viet Nam. The project will not employ any form of forced or compulsory labour or involve any child labour.	1. N/A 2. N/A 3. N/A 4. N/A 5. N/A

principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave.		2. Workers have to rights to establish and join labour organisations as per regulated in Labour Code. 3. The project owner will follow Labour Code of Viet Nam. All workers will have labour contracts with contents as required. 4. The project activity does not involve any child labour. 5. Workers will participate in the operation training before the completion of the plant.	
---	--	--	--

4. No child labour is allowed (Exceptions for children working on their families' property requires an <u>Expert Stakeholder</u> opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures			
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	1. No	No potential risks to the local economy. The main investment of the project activity is the initial construction cost of the wind power plant and it will be covered by the project developer. Once commissioned, the project will increase the industrial share in the economic structure of Gia Lai province and lever the performance of traditional trade villages as well as tourism industry and services inside the province.	1. N/A
Principle 7.1 Emissions			

Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The project activity involves in wind based power technology and does not lead any increase in greenhouse gas emissions over the baseline scenario. The project activity will generate emission reductions which is estimated at 147,519 tCO2e/year.	N/A
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project activity supplies energy to national grid and displaces equivalent quantity of electricity which would have been generated by fossil fuel dominated grid connected power plants. The plant generates and supplies electricity to national grid and for internal use. In case of the shutdown of the WTGs, the internal electricity consumption is taken from the national grid and measured by the bi-directional power meters. Therefore, the project does not use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource that provides for	N/A

		other local users.	
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project activity is wind power technology and does not affect the natural or pre-existing pattern of watercourses, ground-water and/ or the watershed(s).	N/A
Principle 8.2 Erosion and/or Water Body Instability			
Could the Project directly or indirectly cause additional erosion and/or water body instability or disrupt the natural pattern of erosion?	No	The project activity is on-shore wind power technology and does not affect erosion and/ or water body stability.	N/A
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project activity is wind power technology and does not involve the use of land and soil for production of crops or other products.	N/A
Principle 9.2 Vulnerability to Natural Disaster			

Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The project is to generate electricity from wind energy. Therefore, the project will not be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions.	N/A
Principle 9.3 Genetic Resources			
Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	The project is to generate electricity from wind energy. Therefore, the project will not negatively impacted by the use of GMOs.	N/A
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	Construction phase: <i>Gas emission, dust:</i> The project is a wind power project, so the impact on air quality is only temporary during construction period and not significant. Besides, the Project owner has applied the following measures: - Mean/vehicles don't carry	N/A

		materials over the specified deadweight tonnage, use fuel as designed. - Equip labour protection for workers. - Covering means/vehicles, which loads construction materials in order to minimize dust dispersion. - Spraying water along the construction roads and construction site. - Checking and maintaining means/vehicles frequently. <u>Wastewater:</u> Domestic wastewater generated mainly from living activities. For reducing the quantity of domestic wastewater and mitigation of pollution, the following mitigation measures are implemented: - Prioritize recruiting local worker. - Arrange human resources reasonably in construction period. - The wastewater will be collected in temporary toilets or portable toilets, and treated periodically. <u>Solid waste</u>	
--	--	---	--

		- Construction garbage: The waste is separated and reduced. Utilizing most of scraps for construction activities. - Domestic solid waste: To minimize pollution due to solid waste, the project owner conduct measures as follows: + Prioritize recruiting local employees. + Establish rules on hygiene and environmental protection. + Set up rubbish bins to gather waste. + Cooperate with function unit to transport and treat waste as regulated. Operation phase: <u>Gas emission, dust</u> The proposed project operates using wind energy and does not use fossil fuels. Therefore, there is no gas pollutants emitted into the atmosphere during the operation. <u>Wastewater</u> Domestic wastewater will be treated in septic tank, then move to general drainage system. The treated effluent will meet Viet	
--	--	--	--

		Nam Standard before discharging into the environment. <u>Solid waste</u> Domestic solid waste will be gathered in plastic bins and then transported and treated by function unit. <u>Noise</u> Turbine movement in operation phase may emit noise surrounding environment. However, no residents live near to the project site, thus no impacts to settlement. For operation workers, the following measures will be implemented to minimize negative impacts: - Building a solid structural foundation column, resistant to vibrations and shocks. Installing the equipment as per manufacturer's guideline. - Automate the operation process to limit the work time of operation workers. - Arrange work and rest time reasonably to reduce the expose with noise. - Plant tree surrounding the plant <u>Flickering effect</u> When the sun shines on turbine,	
--	--	--	--

		they create a flickering effect and a shaded area in the behind of turbine. However, this impact is very small and only localized. Regarding large turbines of HBRE Chu Prong wind farm, the rotation speed of the propeller is quite low, so the flickering effect is almost non-existent and will not affect the environment or settlement. Moreover, the residential area is far from project site (i.e the nearest distance of 2 km). Therefore, noise and flickering effect doesn't impact to residential area. <i>Ref: EIA Report, page 13, 117, 118, 121, 122</i>	
Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?	Yes	The project is to generate electricity from wind energy. Therefore, the project will not involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials.	Hazardous wastes (waste oil from maintenance of transformer included) if any are collected and treated by the functional third party in accordance with local laws and related regulation. Non- hazardous waste and/or materials (plastic waste included) are collected separately and treated by the functional third

			party in accordance with local laws and related regulation.
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?	No	The project is to generate electricity from wind energy. Therefore, the project will not involve the application of pesticides and/or fertilisers.	N/A
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?	No	The project is a wind power project. It will not involve the harvesting of forests.	N/A
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?	No	The project is to generate electricity from wind energy. Therefore, the project will not modify the quantity or nutritional quality of food available.	N/A
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?	No	The project is to generate electricity from wind energy. Therefore, the project will not involve animal husbandry.	N/A
Principle 9.10 High Conservation Value Areas and Critical Habitats			

Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?	No	The project is not located in an area within a high conservation value area or within critical natural habitats. According to ecological investigation result in implementation of EIA, the project, the project site is mainly rubber plantation land, species of creatures are poor and exhausted with wild shrubs, grass, insects, some small birds. The project site is also no migration path of birds. Therefore, no bird and animal will be harmed in operation process of the project activity. <i>Ref: EIA Report, page 73, 121</i>	N/A
Principle 9.11 Endangered Species			
Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? AND/OR Does the Project potentially impact other areas where endangered species may be present through transboundary affects?	No	There are no any endangered species identified at project site and also no species have the route through area. The project activity does not impact other endangered species through transboundary affects. <i>Refer. EIA Report, page 73</i>	N/A

APPENDIX 2- CONTACT INFORMATION OF PROJECT PARTICIPANTS

Organization name	Coral Future Pte. Ltd
Registration number with relevant authority	-
Street/P.O. Box	20A, Tanjong Pagar Road
Building	-
City	Singapore
State/Region	-
Postcode	088443
Country	Singapore
Telephone	-
E-mail	santosh@coralfuture.com
Website	Coralfuture.com
Contact person	Santosh Kumar Singh
Title	Director
Salutation	Mr.
Last name	Singh
Middle name	Kumar
First name	Santosh
Department	-
Mobile	+1 917 794 1009
Direct tel.	-
Personal e-mail	-

Organization name	HBRE Gia Lai Wind Power Joint Stock Company
Registration number with relevant authority	-

Street/P.O. Box	282 Hung Vuong street, Chu Prong town, Chu Prong district, Gia Lai province, Viet Nam
Building	-
City	Gia Lai province
State/Region	Gia Lai province
Postcode	-
Country	Viet Nam
Telephone	-
E-mail	tuvannangluongxanh@gmail.com
Website	-
Contact person	Supa Waisayarat
Title	Director of region
Salutation	Mr.
Last name	Waisayarat
Middle name	-
First name	Supa
Department	-
Mobile	+84 979 728 019
Direct tel.	-
Personal e-mail	-

APPENDIX 3- LUF ADDITIONAL INFORMATION

Not applicable

APPENDIX 4-SUMMARY OF APPROVED DESIGN CHANGES

Not applicable

Revision History

Version	Date	Remarks
1.2	14 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.1	24 August 2017	Updated to include section A.8 on 'gender sensitive' requirements
1.0	10 July 2017	Initial adoption